



We represent the masses in a spectrum as a sequence *Spectrum* of integers $s_1, \dots, s_m$ in increasing order, where s_1 is zero and s_m is the total mass of the (unknown) peptide. We define a labeled graph $Graph(Spectrum)$ by forming a node for each element of *Spectrum*, then connecting nodes s_i and s_j by a directed edge labeled by an amino acid a if $s_j - s_i$ is equal to the mass of a . As we assumed when sequencing antibiotics, we do not distinguish between amino acids having the same integer masses (i.e., the pairs K/Q and I/L).

The integer mass table is reproduced below:

G	A	S	P	V	T	C	I	L	N	D	K	Q	E	M	H	F	R	Y	W
57	71	87	97	99	101	103	113	113	114	115	128	128	129	131	137	147	156	163	186





CODE CHALLENGE: Construct the graph of a spectrum.

Given: A space-delimited list of integers *Spectrum*.

Return: *Graph(Spectrum)*.

Note: Throughout this chapter, all dataset problems implicitly use the standard integer-valued mass table for the regular twenty amino acids. Examples sometimes use the toy amino acid alphabet {X, Z} whose masses are 4 and 5, respectively.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/graph_spectrum.txt)

Sample Input:

```
57 71 154 185 301 332 415 429 486
```

Sample Output:

```
0->57:G
0->71:A
57->154:P
57->185:K
71->185:N
154->301:F
185->332:F
301->415:N
301->429:K
332->429:P
415->486:A
429->486:G
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11813/step/2



Given an amino acid string *Peptide*, its ideal spectrum, denoted $IdealSpectrum(Peptide)$, is the collection of integer masses of all its prefixes and suffixes. Note that an ideal spectrum may have repeated masses; for example, $IdealSpectrum(GPG) = \{0, 57, 57, 154, 154, 211\}$. We say that an amino acid string *Peptide* **explains** a collection of integers *Spectrum* if $IdealSpectrum(Peptide) = Spectrum$.

Decoding an Ideal Spectrum Problem: *Reconstruct a peptide from its ideal spectrum.*

Input: A collection of integers *Spectrum*.

Output: An amino acid string *Peptide* that explains *Spectrum*.

The following pseudocode solves the Decoding an Ideal Spectrum Problem.

DecodingIdealSpectrum(*Spectrum*)

 construct *Graph*(*Spectrum*)

for each path *Path* from *source* to *sink* in *Graph*(*Spectrum*)

Peptide $\leftarrow$ the amino acid string spelled by the edge labels of *Path*

if $IdealSpectrum(Peptide) = Spectrum$

return *Peptide*





CODE CHALLENGE: Solve the Decoding an Ideal Spectrum Problem.

Given: A space-delimited list of integers *Spectrum*.

Return: An amino acid string that explains *Spectrum*.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/decoding_ideal_spectrum.txt)

Sample Input:

57 71 154 185 301 332 415 429 486

Sample Output:

GPFNA

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11813/step/4



Given an amino acid string $Peptide = a_1 \dots a_n$ of length n , we will represent its prefix masses using a binary **peptide vector** $Peptide'$ with $mass(Peptide)$ coordinates. This vector contains a 1 at each of the n **prefix coordinates**

$$mass(a_1), mass(a_1 a_2), \dots, mass(a_1 a_2 \dots a_n),$$

and it contains a 0 in each of the remaining **noise coordinates**. The toy peptide XZZXX, whose prefix masses are 4, 9, 14, 18, and 22, corresponds to the peptide vector (0,0,0,1,0,0,0,0,1,0,0,0,0,1,0,0,0,1,0,0,0,1) of length 22.

Converting a Peptide into a Peptide Vector Problem: *Convert a peptide into a peptide vector.*

Input: An amino acid string $Peptide$.

Output: The peptide vector $Peptide'$.





CODE CHALLENGE: Solve the Converting a Peptide into a Peptide Vector Problem.

Given: An amino acid string P .

Return: The peptide vector of P (in the form of space-separated integers).

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradatasets/proteomics/peptide_to_peptide_vector.txt)

Sample Input:

XZZXX

Sample Output:

0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 1 0 0 0 1

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11813/step/6



We can also invert the conversion of a peptide into a peptide vector, so that the terms "peptide" and "peptide vector" can be used interchangeably.

Converting a Peptide Vector into a Peptide Problem: *Convert a peptide vector into a peptide.*

Input: A binary vector P .

Output: A peptide whose peptide vector is equal to P (if such a peptide exists).





CODE CHALLENGE: Solve the Converting a Peptide Vector into a Peptide Problem.

Given: A space-delimited binary vector P

Return: An amino acid string whose binary peptide vector matches P . For masses with more than one amino acid, any choice may be used.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradata/proteomics/peptide_vector_to_peptide.txt)

Sample Input:

```
0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 1 0 0 0 1
```

Sample Output:

```
XZZXX
```

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11813/step/8



Given a spectral vector $Spectrum'$, our goal is to find a peptide $Peptide$ maximizing $Score(Peptide, Spectrum')$. Since the mass of a peptide and the parent mass of the spectrum that it generates should be the same, a peptide vector should have the same length as the spectral vector under consideration. We will therefore define the score between a peptide vector and a spectral vector of different length as $-\infty$.

Peptide Sequencing Problem: *Given a spectral vector, find a peptide with maximum score against this spectrum.*

Input: A spectral vector $Spectrum'$.

Output: An amino acid string $Peptide$ that maximizes $Score(Peptide', Spectrum')$ among all possible amino acid strings.

Given a spectral vector $Spectrum' = (s_1, \dots, s_m)$, we will construct a DAG on $m + 1$ nodes, labeled with the integers from 0 (*source*) to m (*sink*), and then connect node i to node j by a directed edge if $j - i$ is equal to the mass of an amino acid. We will further assign weight s_i to node i (for $1 \leq i \leq m$) and assign weight zero to node 0.

Any path connecting *source* to *sink* in this DAG corresponds to an amino acid string $Peptide$, and the total weight of nodes on this path is equal to $Score(Peptide', Spectrum')$. We have therefore reduced the Peptide Sequencing Problem to the problem of finding a maximum-weight path from source to sink in a node-weighted DAG.



CODE CHALLENGE: Solve the Peptide Sequencing Problem.

Given: A space-delimited spectral vector *Spectrum'*.

Return: An amino acid string with maximum score against *Spectrum'*. For masses with more than one amino acid, any choice may be used.

Note: When a spectral vector $Spectrum' = s_1 \dots s_m$ is given, it does not have a zero-th element; in your implementations, you should assume that s_0 is equal to zero.

Extra Dataset (http://bioinformaticsalgorithms.com/data/extradata/sets/proteomics/peptide_sequencing.txt)

Sample Input:

0 0 0 4 -2 -3 -1 -7 6 5 3 2 1 9 3 -8 0 3 1 2 1 8

Sample Output:

XZZXX

Time Limit: 5 mins

To solve this problem please visit stepic.org/lesson/-11813/step/10